

Toby Nguyen (Nguyễn Đình Tú)

Technical Lead | Senior Software Engineer | Product-Minded Technologist

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Location: Ha Dong, Ha Noi, Vietnam | Date of Birth: 03/1993

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Technical Lead & Senior Software Engineer with a **Product-first mindset**, specialized in **distributed backend systems, high-scale web applications, and optimizing engineering productivity through AI-augmented workflows (MCP, LLMs)**.

CAREER SUMMARY

Accomplished Technical Lead & Senior Software Engineer with over 8+ years of experience engineering high-scale distributed systems, resilient backend services, and world-class web applications. Deeply versatile across systems programming (Rust, Java/Spring Boot, Node.js) and modern web ecosystems (React, Next.js, Vue.js, Micro-frontends). Proven expertise in data pipelines, real-time telemetry streaming (WebSockets, SSE), distributed caching, and Core Web Vitals optimization.

Pioneer in marrying a **Product-Led Growth (PLG)** mindset with **AI-augmented engineering workflows (MCP, LLM toolchains)** to drive technical excellence, deliver resilient software architectures, and scale cross-functional engineering teams.

CORE COMPETENCIES & TECHNICAL SKILLS

Category	Key Expertise & Technologies
Backend & Systems Engineering	Java, Spring Boot (Spring Data JPA, Spring Security), Rust (Tokio Async, Win32 Job Objects, Concurrency), Node.js (NestJS, Express), TypeScript server-side, Event-Driven Architecture
API Design & Real-Time Streaming	RESTful APIs, GraphQL, gRPC Contracts, WebSockets, Server-Sent Events (SSE), API Gateways, Backend-for-Frontend (BFF), Distributed IPC
Databases & Distributed Caching	PostgreSQL (Supabase), MS SQL Server, MySQL, Redis (Pub/Sub, Caching Strategies, Session Management), ClickHouse (OLAP), Apache Solr, SQLite, Schema Design & Index Optimization
Frontend Architecture & Web	React Ecosystem (Concurrent Features, Next.js, Server Components), Vue.js Ecosystem (Composition API, NuxtJS), TypeScript, Angular, Cross-Platform Desktop (Electron JS)
Scalability & Performance	Micro-frontends (Module Federation), Component-Driven Design Systems (Storybook, Radix UI), Core Web Vitals (LCP < 2.0s, INP < 150ms, CLS ≈ 0), Critical Rendering Path, Dynamic Code Splitting

Category	Key Expertise & Technologies
Real-time Data Visualization	High-Density Canvas/SVG Charting, Virtual Scrolling / Grids, ECharts, Highcharts, Optimistic UI, Client-side Caching (TanStack Query, SWR)
Cloud, DevOps & Infrastructure	Docker, Docker Compose, Kubernetes fundamentals, Azure Cloud, Linux CLI, Bash/PowerShell, GitHub Actions CI/CD, Monorepo Orchestration (Turborepo, Nx)
AI-Augmented Engineering	Model Context Protocol (MCP), LLM-assisted Software Engineering, Prompt Engineering for Toolchains, Automated Testing Suites (Playwright, Cypress, Vitest, Jest)
Engineering Leadership	Technical Strategy, Cross-Squad Engineering Leadership (15+ engineers), System Architecture Roadmap, Technical Debt Governance, Agile/Scrum Orchestration

Skills Assessment Table

Skill Area	Focus	Notes
Backend & Systems Design	Microservices, Runtimes	Java / Spring Boot, Rust async (Tokio), Node.js/NestJS, Win32 Job Objects, Event-Driven Architecture
APIs & Streaming	Real-time, Communication	WebSockets telemetry feeds, SSE streaming, RESTful, GraphQL, gRPC, BFF orchestration
Databases & Caching	Data Storage, Performance	PostgreSQL, MS SQL, Redis caching strategies, ClickHouse OLAP, query and index optimization
Frontend Architecture	Micro-frontends, Systems	Micro-frontends (Module Federation), Enterprise Design Systems, Monorepo (Turborepo), Clean Code
Web Performance Tuning	Core Web Vitals, Rendering	Sub-2s LCP, INP < 150ms, zero layout shifts, dynamic code splitting, bundle size minimization
Modern Web Ecosystem	TypeScript, React, Vue	Deep expertise in React Ecosystem, Next.js, Vue.js Composition API, NuxtJS, Tailwind CSS, Electron JS
Real-time Data Visualization	Streaming, High-Density Charts	Real-time WebSockets telemetry, Virtual Grids, ECharts, Canvas/SVG, Client-side Caching (SWR)
DevOps & Cloud	Containers, CI/CD	Docker, Azure Cloud, Linux CLI, GitHub Actions CI/CD, process supervision & multi-runtime tools
AI-Augmented Development	Productivity, Automation	Model Context Protocol (MCP), GenAI-assisted engineering, LLM toolchains, Playwright E2E

PROFESSIONAL EXPERIENCE

Company Name	Position	Duration
CMC GLOBAL	Technical Project Lead / Tech Lead	9/2024 – Present
ICOMM TECH.JSC	Chief Product Officer (CPO) / Tech Lead	5/2020 – 8/2024
OPEN COMMERCE GROUP	Senior Software Engineer	12/2018 – 4/2020
TDT ASIA JSC	Fullstack Software Engineer	5/2018 – 12/2018
LUSH ERA LLC	Software Engineer / Web Developer	12/2016 – 4/2018

FEATURED PROJECT LIST

(09/2024 – Present) Lotte World Ecosystem – Theme Park Web & Mobile Portals

- **Timeline:** 09/2024 – Present
- **Company:** CMC GLOBAL
- **Client:** Lotte World (Theme Parks, Entertainment & Hospitality)
- **Project description** *: Large-scale multi-runtime enterprise ecosystem for Lotte World amusement parks (Jamsil & Busan), encompassing high-traffic web & webview portals, real-time attraction wait-time & reservation engines, and digital interactive park navigation.
- **Team size:** 15+ engineers
- **Responsibilities** *:
 - Participated as **Core Frontend Maintainer**, maintaining, refactoring, and delivering feature enhancements for Park Visitor Web Portals and Partner Booking WebViews strictly adhering to client functional specifications and technical docs.
 - Implemented responsive Web and mobile-embedded WebView user interfaces using React, Vite, and Tailwind CSS, ensuring UI fidelity with design documentation and smooth interactions.
 - Configured multi-runtime local developer workflows and mock data synchronization to streamline testing between Spring Boot backend services and React/React Native frontend clients.
 - Performed continuous code maintenance, bug fixes, and performance tuning to guarantee cross-browser compatibility and responsive UX across mobile WebViews.
- **Programming Language** *: TypeScript, JavaScript, Java
- **Framework** *: React, Vite, React Native, Tailwind CSS, Spring Boot
- **Platform, server and database** *: Docker, Redis, MS SQL Server, Linux
- **Methodology:** Agile / Scrum

(09/2024 – Present) Digital Twin – Electric Vehicle (EV) Real-Time Telemetry Dashboard

- **Timeline:** 09/2024 – Present
- **Company:** CMC GLOBAL
- **Client:** Leading Automotive Manufacturer (Automotive Tech, IoT Integration)
- **Team size:** 15+ engineers (3 squads)

- **Project description** *: Large-scale real-time Digital Twin telemetry dashboard platform for aggregating, monitoring, and analyzing telemetry and operational status for fleets of electric vehicles (EVs).
- **Responsibilities** *:
 - Served as **Technical Project Leader & Tech Lead**, establishing overall technical architecture, engineering standards, and real-time telemetry streaming across 3 squads (15+ engineers) processing millions of daily EV operational events.
 - Architected a **Micro-frontends** architecture and standardized a shared enterprise **Design System**, ensuring visual consistency and decoupled release cycles across squads.
 - Optimized real-time telemetry streaming via WebSockets by implementing **Virtual Scrolling, Canvas Data Charting, and Debounced/Throttled State Updates**, sustaining a continuous 60fps rendering rate under high telemetry volume without blocking the Main Thread.
 - Formulated a comprehensive **Core Web Vitals** strategy: orchestrated route-based and component-based dynamic code splitting, reducing First Input Delay / INP to < 150ms and maintaining LCP < 2.0s across all mission-critical monitor views.
 - Introduced **AI-augmented Frontend SDLC** practices (LLM UI generation, automated visual regression testing), elevating frontend feature delivery velocity by 25%.
- **Programming Language** *: TypeScript, JavaScript, HTML5, CSS3, Rust (BFF tooling)
- **Framework** *: React (Concurrent Features), Next.js, Tailwind CSS, Zustand, Node.js
- **Platform, server and database** *: Azure Cloud, Docker, Kubernetes, WebSocket, PostgreSQL, Redis
- **Methodology**: Agile / Scrum

(05/2020 – 08/2024) Big Data Analytics & Interactive Intelligence Platform

- **Timeline**: 05/2020 – 08/2024
- **Company**: ICOMM TECH.JSC
- **Client**: Government, Banking & Aviation Sectors
- **Team size**: 15+ engineers
- **Project description** *: Multi-platform enterprise Big Data extraction and intelligence dashboard, delivering interactive data visualization, real-time reporting, and deep analytics under strict enterprise security constraints.
- **Responsibilities** *:
 - Directed product roadmap and technical architecture as **Chief Product Officer (CPO) & Technical Lead**.
 - Engineered an interactive **Data Visualization Dashboard** powered by ECharts and Highcharts, solving high-density data rendering challenges for 100,000+ complex visual data points without frame drops.
 - Implemented **Client-side Caching (SWR/React Query patterns)** and **Virtualized Data Grids**, enabling near-instantaneous search, filtering, and paging across millions of records.
 - Established automated frontend quality assurance suites (End-to-End testing with Cypress/Playwright and unit testing with Jest), achieving 99.9% application uptime SLA.
 - Mentored 15+ engineers in Component-Driven Architecture, JavaScript Clean Code principles, and runtime render profiling.
- **Programming Language** *: TypeScript, JavaScript, HTML5, SCSS/CSS3, Python

- **Framework** *: React Ecosystem, Redux Toolkit, ECharts, .NET Core (Backend API)
- **Platform, server and database** *: Docker, Kubernetes, Redis, Apache Solr, ClickHouse, Linux
- **Methodology**: Agile / Scrum

(12/2018 – 04/2020) High-Converting E-Commerce Storefront & Merchant Suite

- **Timeline**: 12/2018 – 04/2020
- **Company**: OPEN COMMERCE GROUP
- **Client**: Global E-commerce Merchants
- **Team size**: 8–10 members
- **Project description** *: Ultra-fast e-commerce storefront platform and intelligent cross-selling engine integrated across Shopify, WooCommerce, and WordPress ecosystems.
- **Responsibilities** *:
 - Served as **Senior Software Engineer**, spearheading storefront user interface development and AI-driven cross-selling recommendation widgets geared toward Product-Led Growth (PLG).
 - Maximized conversion rate optimization (CRO) and checkout velocity: drove **First Contentful Paint (FCP) to < 1.2s** and **LCP to < 1.8s**, contributing to an estimated 15–20% increase in merchant revenue.
 - Engineered reusable, modular UI components using **Vue.js and React**, enforcing scoped styles, responsive image optimization (WebP/AVIF), and aggressive bundle size minimization.
 - Delivered seamless Mobile-First responsive layouts and frictionless multi-gateway payment flows.
- **Programming Language** *: TypeScript, JavaScript, HTML5, CSS3, Rust (microservices)
- **Framework** *: Vue.js (Vuex), React, Tailwind CSS, Laravel (Backend API)
- **Platform, server and database** *: AWS, Nginx, Redis, MySQL
- **Methodology**: Agile / Scrum

(05/2018 – 12/2018) Enterprise Japanese Business Management Portal

- **Timeline**: 05/2018 – 12/2018
- **Company**: TDT ASIA JSC
- **Client**: Japanese Small & Medium Enterprises (SMB Solutions – Japan Market)
- **Team size**: 6–8 members
- **Project description** *: Enterprise Resource Planning (ERP) and business management web portal customized to meet the rigorous quality and UX standards of the Japanese corporate market.
- **Responsibilities** *:
 - Operated as **Fullstack Software Engineer**, taking primary ownership of the Single Page Application (SPA) client architecture.
 - Leveraged **Angular with RxJS Observables** for reactive state handling and orchestrated complex business workflows with Angular Reactive Forms.
 - Ensured strict **Cross-Browser Compatibility** (Chrome, Firefox, Safari, Edge) and achieved pixel-perfect adherence to Japanese UX/UI specifications.
- **Programming Language** *: TypeScript, JavaScript, HTML5, SCSS
- **Framework** *: Angular, Angular Material, RxJS, Spring Boot (Backend API)

- **Platform, server and database** *: Tomcat, Docker, MySQL, Linux
- **Methodology**: Agile / Scrum

(12/2016 – 04/2018) Responsive Real-time Telemedicine Web Application

- **Timeline**: 12/2016 – 04/2018
- **Company**: LUSH ERA LLC
- **Client**: Medical Clinics & Healthcare Service Providers
- **Team size**: 5 members
- **Project description** *: Secure real-time telemedicine and remote healthcare consultation web application connecting patients with certified medical specialists via video and chat.
- **Responsibilities** *:
 - Acted as **Software Engineer / Web Developer**, building an intuitive, accessible, and fast-loading telemedicine web client.
 - Engineered **Mobile-First Responsive Layouts**, prioritizing Web Accessibility (a11y) standards for elderly patients and individuals with visual impairments.
 - Implemented interactive WebRTC video calling interfaces and low-latency real-time messaging UI.
- **Programming Language** *: JavaScript, HTML5, CSS3, PHP
- **Framework** *: jQuery, Bootstrap, Laravel (Backend)
- **Platform, server and database** *: MySQL, Apache, Ubuntu Linux
- **Methodology**: Agile

(2024 – Present) Tuquet – Developer CLI & Distributed Automation Ecosystem

- **Timeline**: 2024 – Present
- **Company**: Independent Product Labs (Open-Source R&D)
- **Client**: Self-initiated Open-Source Developer Tooling & Global Automation Engineers
- **Team size**: Creator & Principal Architect (Solo / Core Team)
- **Project description** *: High-performance open-source developer tooling ecosystem and distributed browser automation suite consisting of a unified master CLI (**tuquet**) in Rust, a visual workflow web studio in Vue.js, a supervised background runner daemon (**tuquet runner**), and cloud coordination services.
- **Responsibilities** *:
 - Acted as **Creator & Principal Architect**, architecting the end-to-end multi-runtime ecosystem spanning systems programming in Rust, visual web studio in Vue.js, and distributed task supervision.
 - Engineered the master CLI (**tuquet**) in **Rust**: built an interactive scoped shell with hierarchical state management, rustyline-based smart tab-completion, ANSI color styling, and sub-command delegation (**automa**, **runner**, **cloud**, **browser**).
 - Developed the visual automation workflow studio with **Vue.js Ecosystem & TypeScript**: designed an interactive node-graph canvas (Drawflow), custom automation blocks, and a real-time execution inspector with instant visual feedback.
 - Architected a zero-leakage process supervision engine in the runner daemon (**tuquet runner**) leveraging Win32 Job Objects to eliminate orphan Chromium zombie processes and strictly isolate CPU/RAM.

- Engineered a high-throughput Medallion data crawler pipeline (Bronze gzip BLOB storage -> Silver Rust deduplication & HTML sanitizer -> Gold Supabase synchronization).
 - Standardized local workspace configurations into a Single Source of Truth (~/.tuquet/), eliminating configuration drift across CLI, extensions, and runner runtimes.
 - Automated binary builds, multi-platform packaging, and distribution through an official Windows Scoop bucket (tuquet/scoop-bucket).
 - **Programming Language** *: Rust, TypeScript, JavaScript, SQL
 - **Framework** *: Vue.js Ecosystem, Tailwind CSS, Drawflow, Tokio (Rust Async), Node.js
 - **Platform, server and database** *: Supabase, PostgreSQL, WebSockets, Win32 Job Objects, Docker, Scoop, GitHub Actions
 - **Methodology**: Agile / Kanban
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ACHIEVEMENTS & AWARDS

- **Q4/2025: Rising Star Award (CMC GLOBAL)** – Recognized for exemplary technical leadership and high-impact delivery of the Digital Twin EV project.
 - **2023: Team of the Year (ICOMM TECH.JSC)** – Leading the engineering organization to exceed all product delivery targets.
 - **2022: Outstanding Employee (ICOMM TECH.JSC)** – Awarded for spearheading the flagship Big Data platform.
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EDUCATION

FPT University | Information Technology (1/2012 – 9/2014)

- **Focus**: Full-Stack Web Development, Desktop Application Design & Software Architecture.
- **Academic Standing**: GPA 8.2 / 10
- **English Proficiency**: TOEIC 650 (Fluent technical reading/writing & professional collaboration)